Typing Odia Phonetically

A reference sheet: what to type in English to get ଓଡ଼ିଆ

v2 — with Sanskrit in Odia script

Draft v0.2 · July 29, 2026

Pitfall. This sheet is a **draft scheme**, derived from ITRANS conventions and published Odia orthography references. It has **not been reviewed by a native Odia speaker**. Treat every mapping as a proposal, not a fact.

Note. What v2 adds. Everything in v1 still holds — this document is a superset, so nothing you already knew has changed. Sections 6 onward cover Sanskrit written in Odia script: the register of stotras, Puranas, Upanishads and Jagannatha devotional verse. Those additions live in `Appv2/rules/*.delta.toml` and never touch `core/rules/symbols.toml`, which remains the single source of truth for everyday Odia.

1 How it works

Odia is an *abugida*, not an alphabet. That one fact explains most of the rules below.

Key idea. Every consonant already carries a built-in vowel. କ on its own is `ka`, not `k`. To attach a different vowel you add a **matra** (a vowel sign). To attach *no* vowel — so the consonant joins the next one — the script uses a **halant**, which produces a conjunct.

There are two typing modes.

Natural mode (default)
: Type it the way it sounds. Case is ignored. Several spellings reach the same word, and the engine offers a ranked candidate list — you pick. This is what you will use day to day.

Strict mode (`--strict`)
: Full ITRANS. Case matters. Exactly one deterministic answer, no candidates. Use it when you need a specific character and do not want to argue with the ranking.

And, in v2, three *ranking* profiles

Orthogonal to the two modes above. These do not change what is *reachable* — every profile can produce every character — only what the candidate list offers first.

Everyday
: The original engine, unchanged.

Odia + Sanskrit (default)
: Everyday Odia leads; Sanskrit spellings sit a slot or two down but are always present. Costs nothing in everyday accuracy.

Sloka For verse. A word-final ° beats ି, ଂ beats ଃ, and the Sanskrit vocalics become typeable.

Key idea. The profiles only diverge on words **nothing in the dictionary knows**. For an attested word the lexicon wins in every profile — `balam` is ବଳଂ° in all three, because ବଳଂ° is a real word and a rule guess should not outrank one.

In the tables below, the **Type** column is natural mode and **Strict** is the ITRANS form. Where they are identical, Strict is left blank.

2 Vowels

Each vowel has two shapes: an **independent** letter, used at the start of a word or after another vowel, and a **matra**, used after a consonant. You type the same thing for both — the engine picks the right shape from context.

Independent	On କ	Type	Strict	Example
ଅ	କ	`a`		ଅଛ `achha`
ଆ	କା	`aa`	`A`	ଆମ `aama`
ଇ	କି	`i`		ଇଏ `ie`
ଈ	କୀ	`ii`, `ee`	`I`	ନଈ `naii`
ଉ	କୁ	`u`		ଉଠ `utha`
ଊ	କୂ	`uu`, `oo`	`U`	ମୂଳ `muula`
ଋ	କୃ	`ru`, `ri`	`R^i`	ଋଷି `rushi`
ଏ	େକ	`e`		ଏକ `eka`
ଐ	ୈକ	`ai`, `oi`		ଐକ୍ୟ `aikya`
ଓ	ୋକ	`o`		ଓଠ `otha`
ଔ	ୌକ	`au`, `ou`		ଔଷଧ `aushadha`

Pitfall. **Double the vowel to lengthen it.** This is the single most common reason a word comes out wrong. `a` is ଅ but `aa` is ଆ, so ଭାଗ୍ୟ is `bhaagya`, not `bhagya`. Likewise `ii`/`ee` for ଈ and `uu`/`oo` for ଊ. Frequent words are recognised either way, but for anything else the doubling matters.

Note. The rare vowels ୠ `RR`, ଌ `Lu` and ୡ `LL` exist in the scheme but almost never appear in modern Odia. They are available in strict mode only.

3 Consonants

The five *vargas* (articulation groups). Read each row left to right: unaspirated, aspirated, voiced, voiced-aspirated, nasal.

	Position in the mouth				
Group	voiceless		voiced		nasal
Velar	କ `ka`	ଖ `kha`	ଗ `ga`	ଘ `gha`	ଙ `nga`
Palatal	ଚ `cha`	ଛ `chha`	ଜ `ja`	ଝ `jha`	ଞ `nya`
Retroflex	ଟ `Ta`	ଠ `Tha`	ଡ `Da`	ଢ `Dha`	ଣ `Na`
Dental	ତ `ta`	ଥ `tha`	ଦ `da`	ଧ `dha`	ନ `na`
Labial	ପ `pa`	ଫ `pha`	ବ `ba`	ଭ `bha`	ମ `ma`

Pitfall. **Retroflex vs dental** is the one distinction English does not have, and the main reason you will see multiple candidates. Retroflex (ଟ ଠ ଡ ଢ ଣ) is made with the tongue curled back; dental (ତ ଥ ଦ ଧ ନ) with the tongue on the teeth. In strict mode a **capital** letter means retroflex: `Ta` is ଟ, `ta` is ତ. In natural mode case is ignored and both are offered — the dictionary decides which is more likely.

Everything else

Letter	Type	Notes	Letter	Type	Notes
ଯ	`ya`	sounds like *ja*	ଶ	`sha`	palatal
ର	`ra`		ଷ	`Sha`	retroflex
ଲ	`la`		ସ	`sa`	dental
ଳ	`La`	retroflex *l*	ହ	`ha`	
ବ	`ba`, `va`		ୱ	`wa`	ସ୍ୱର `swara`

The three special letters

These carry a *nukta* (a dot below) or are otherwise distinct. They matter more in Odia than in most Indic scripts — the first one appears in the name of the language.

Letter	Type	Sound	Example
ଡ଼	`rr`, `d`	retroflex flap	ଓଡ଼ିଶା `odishaa`, ବଡ଼ `bada`
ଢ଼	`rh`, `dh`	aspirated flap	ପଢ଼ା `paRhaa`
ୟ	`y` (not word-initial)	true *y* sound	ରାଜ୍ୟ `raajya`

Pitfall. ଯ and ୟ are different letters, and Odia uses them the opposite way round from Hindi. ଯ is pronounced like *ja*; ୟ is the one that actually sounds like *y*. So typing `y` gives ଯ at the *start* of a word (`yoga` → ଯୋଗ) and ୟ anywhere else (`raajya` → ରାଜ୍ୟ, `nyaaya` → ନ୍ୟାୟ). Both stay in the candidate list, so you can override either way.

4 Signs and marks

These are combining marks, so they are shown here attached to କ — on their own they would be almost invisible.

Mark	On କ	Name	Type	Example
ଂ	କଂ	anusvara	`n`, `m`, `M`	ଅଂଶ `ansha`
ଁ	କଁ	candrabindu	`n~`, `Mn`	ମୁଁ `mun`
ଃ	କଃ	visarga	`H`	ଦୁଃଖ `duHkha`
◌୍	କ୍	halant	automatic	see below

Anusvara nasalises *before a following consonant*; candrabindu nasalises the vowel itself; visarga is a breathy release; halant strips the built-in vowel so a conjunct can form.

5 Conjuncts

When two consonants meet with no vowel between them, they fuse into a single conjunct glyph. **You do not type the halant** — just type the consonants back to back and the engine forms the conjunct.

Type	Get	Type	Get	Type	Get
kta	କ୍ତ	pra	ପ୍ର	nta	ନ୍ତ
ksha	କ୍ଷ	shra	ଶ୍ର	Nda	ଣ୍ଡ
jny	ଜ୍ଞ	swa	ସ୍ୱ	mbha	ମ୍ଭ
tra	ତ୍ର	nka	ଙ୍କ	ddha	ଦ୍ଧ
sta	ସ୍ତ	ShTa	ଷ୍ଟ	lla	ଲ୍ଲ
gya	ଗ୍ୟ	nya	ନ୍ୟ	dya	ଦ୍ୟ

Key idea. **A typed `n` is five different letters**, depending on what follows it, because a nasal takes the place of articulation of the next consonant: `n+k/g` → ଙ (ଙ୍କ, ଗଙ୍ଗ); `n+ch/j` → ଞ (ପଞ୍ଚମ); `n+T/D` → ଣ (ମୁଣ୍ଡ); `n+t/d` → ନ (ଅନନ୍ତ); `n+p/b` → ମ (ସମ୍ବାଦ). The anusvara ଂ is offered alongside all of them (ସଂସ୍କାର). So the honorific is simply `nka`: `baabunka` → ବାବୁଙ୍କ.

Pitfall. `ny` and `gy` are read as *two* letters, not as ଞ and ଜ୍ଞ, because in real words that is nearly always what they are --- `dhanyabaada` → ଧନ୍ୟବାଦ, `bhaagya` → ଭାଗ୍ୟ. For the single letters type `~n` → ଞ and `jny` → ଜ୍ଞ.

Note. To force a bare consonant with a visible halant and *no* conjunct — which you occasionally want at the end of a word — append an underscore: `kamal_` gives କମଲ୍ rather than କମଲ.

6 Sanskrit in Odia script

The Odia script is used to write Sanskrit as well as Odia — stotras, Puranas, Upanishads, the Bhagavata, Jayadeva's Gita Govinda. That register needs characters and spellings that everyday Odia rarely touches, and everything in this section exists for it.

Four marks you type directly

These four cannot come from the rule tables at all: a roman key must be lowercase `a--z`, and a mark that opens a word is unreachable by construction. So they are substituted before the engine sees the text.

Type	Get	Name	Example
|	।	danda	`balam |` → ବଳଂ ।
||	॥	double danda	`balam ||` → ବଳଂ ॥
'	ଽ	avagraha	`so'ham` → ସୋଽହଂ
:	ଃ	visarga	`du:kha` → ଦୁଃଖ

Note. **Two deliberate asymmetries.**

The apostrophe only fires **after `o` or `e`**, so `don't` and `Krishna's` survive untouched. That is not a hedge: Sanskrit sandhi puts avagraha only after those two vowels, and across every occurrence in an 18.4M token Odia corpus, 98% follow ଓ ା or ଏ.

The colon becomes the *letter* `h`, not ଃ directly. Emitting the Odia character would split the word in two and each half would be transliterated blind — `du:kha` came out ଦୁ ଃ ଖା even though ଦୁଃଖ is in the dictionary. Avagraha has no such problem, because it only ever sits at a seam *between* two words, where splitting is the correct analysis.

The Sanskrit vocalics

Only ୃ is in regular use in modern Odia. The rest exist in the script specifically for Sanskrit transcription, and are reachable by keystroke in Sloka mode or from the insert palette in any mode.

Letter	On କ	Name	Type	Note
ଋ	କୃ	vocalic r	`ri`, `ru`	common in modern Odia
ୠ	କୄ	vocalic rr	`rri`	Sloka mode / palette
ଌ	କୢ	vocalic l	`lri`	Sloka mode / palette
ୡ	କୣ	vocalic ll	`llri`	Sloka mode / palette

Note. These are Sloka-only for a measured reason. The key `rri` sits on top of the existing two-character key `rr` → ଡ଼, which is what made 47 conjuncts reachable in v1. ୠ occurs 22 times in 527k word types; ଡ଼ is in the language's own name. Paying candidate-list noise on ଡ଼ everywhere to reach ୠ is a bad trade outside verse.

Homorganic nasal conjuncts

A nasal before a stop assimilates in the script — ଙ before velars, ଞ before palatals — but nobody types it that way, so a plain `n` reaches all of them. Together these are 3.6% of all Odia text, and ଙ୍କ alone is the third most common conjunct in the language.

Type	Get	Type	Get	Type	Get	Type	Get
`nk`	ଙ୍କ	`nkh`	ଙ୍ଖ	`ng`	ଙ୍ଗ	`ngh`	ଙ୍ଘ
`nch`	ଞ୍ଚ	`nchh`	ଞ୍ଛ	`nj`	ଞ୍ଜ	`njh`	ଞ୍ଝ

Subjoined ୱ

Sanskrit *va* is written ବ standing alone --- ଦେବ, ବିଷ୍ଣୁ --- but ୱ when subjoined under another consonant. This is not a nicety: ସ୍ୱାମୀ outnumbers ସ୍ୱାମୀ 3.3 to 1 in real Odia text, and ବିଶ୍ୱ beats ବିଶ୍ୱ by 7.7 to 1.

Type	Get	Type	Get	Type	Get	Type	Get
`sv`, `sw`	ସ୍ୱ	`dv`, `dw`	ଦ୍ୱ	`tv`, `tw`	ତ୍ୱ	`shv`	ଶ୍ୱ

Note. These are spelled out as their own keys because the rule language has no context meaning "immediately after a halant". The available contexts are word-initial, non-initial, intervocalic and word-final, and all are evaluated against the *roman* you typed, not against the Odia being assembled. Adding a post-halant context is the one engine change worth making later.

ଳ against ଲ, measured

Sanskrit *la* in an Odia tatsama word is usually ଳ. This is the single most visible difference between v1 and v2 output, so it is worth stating what it rests on. Counting every word type in an 18.4M token corpus, ଳ against ଲ for the same skeleton:

Position	Ratio	Prefers
word-final	2.37	ଳ
intervocalic	1.37	ଳ
in a conjunct	1.21	ଳ
subjoined	0.23	ଲ
word-initial	0.05	ଲ

Individual words are far more lopsided than the aggregate — ଜଳ 7.7×, ଫଳ 9.3×, ବଳ 3.6× — but the aggregate is the number the weights answer to.

Pitfall. "Word-final" here means what the *engine* means: nothing follows in the roman. ଥିଲା is typed `thilaa`, so its ଲ is **intervocalic**, not word-final. Measuring it the other way sweeps every -ଲା verb form into the wrong bucket and drags this ratio from 2.37 down to 0.99 — which would justify exactly the opposite weight.

The insert palette

Some marks are too rare to deserve a keystroke and would only add noise to the candidate list. Those live on a tap-to-insert strip above the output: the vocalics above, ଂ ଃ ଽ, the halant and nukta, bare ଋ ଢ଼ ୟ, the two danda, the pranava ଓଁ, and the Vedic accents U+0951 *udatta* and U+0952 *anudatta* for swara-marked chanting text.

Bare ଋ is palette-only by design. Giving it a `ny` key was tried and reverted: it made the correct ନ୍ୟ conjunct pay the digraph-split penalty and dropped `dhanyabaada` out of the candidate list entirely.

Note. Deliberately not included, so nobody re-adds them. U+0B55 ORIYA SIGN OVERLINE reads like a Vedic mark and is not one — Unicode added it in 13.0 for the **Kui** language. U+0B56 and U+0B57 are pieces of a decomposed ଓ◌ି/ଓ◌ୀ, which Unicode says to write precomposed. U+0B72–U+0B77 are fraction signs. Odia has no dedicated OM character; the pranava is ଓ plus candrabindu.

7 Numerals

Type	0	1	2	3	4	5	6	7	8	9
Get	୦	୧	୨	୩	୪	୫	୬	୭	୮	୯

Odia numerals are **off by default** — modern written Odia overwhelmingly uses Arabic digits. Turn them on with the `digits = "odia"` setting.

8 Worked examples

Type this	Get this	Meaning
`namaskaara`	ନମସ୍କାର	hello
`kemiti achha`	କେମିତି ଅଛ	how are you
`dhanyabaada`	ଧନ୍ୟବାଦ	thank you
`mun bhala achhi`	ମୁଁ ଭଲ ଅଛି	I am well
`odishaa`	ଓଡ଼ିଶା	Odisha
`odiaa`	ଓଡ଼ିଆ	Odia (the language)
`bhubaneswara`	ଭୁବନେଶ୍ୱର	Bhubaneswar
`paani`	ପାଣି	water
`tume`	ତୁମେ	you
`bhaagya`	ଭାଗ୍ୟ	fate, luck
`raajya`	ରାଜ୍ୟ	state
`nyaaya`	ନ୍ୟାୟ	justice

Key idea. You are not expected to spell these correctly on the first try. The engine offers a ranked list, you choose, and it remembers your choice. Accuracy on the words *you* personally use climbs quickly — that is the whole design.

9 Five things that trip people up

1. **Double the vowel to lengthen it.** By far the most common mistake. ଭାଗ୍ୟ is `bhaagya`; `bhagya` would be ଭଗ୍ୟ, which is not a word. Same for `ii`/`ee` and `uu`/`oo`.

2. **The built-in vowel is not silent.** ଭଲ is `bhala`, not `bhal`. It is also pronounced closer to *o* than *a* — which is why ଓଡ଼ିଶା is written "Odisha" in English, not "Adisha".

3. **ଡ଼ is not ଡ.** The flap ଡ଼ is extremely common in Odia (ବଡ଼, ପଢ଼ା, ଓଡ଼ିଶା). Internally it is always stored as ଡ plus a separate nukta character, never as a single precomposed character — they look identical on screen but do not match each other in a search.

4. **ଳ is a real letter,** not a typo for ଲ. Retroflex *l*, common in Odia: ମୂଳ, ଫଳ, କାଳ. Type `La`.

5. **Case only matters in strict mode.** In natural mode `Ta` and `ta` are the same input and you disambiguate by picking a candidate.

10 Fixing this sheet

Every mapping above corresponds to an entry in `core/rules/symbols.toml`. When a common word comes out wrong, adjust the weight there rather than working around it:

```
[[consonant]]
roman = "s"
odia  = [
    { text = "ସ", w = 1.0 },
    { text = "ଶ", w = 0.5 },
    { text = "ଷ", w = 0.4 },
]
```

Raising a weight promotes that character in the candidate ranking. After any change, run `npm run`

`eval` and confirm the accuracy delta against the committed baseline before keeping it — a fix for one word often costs you three others.

The Sanskrit mappings are a *delta*

Everything in section 6 lives in `Appv2/rules/sanskrit.delta.toml` and `unified.delta.toml`, which are applied to `symbols.toml` by a compiler. Nothing in `core/` is edited. Each delta block names an id, the weight, and the counts that justify it:

```
[[consonant.replace]]
id    = "S2"
roman = "l"
odia  = [
    { text = "ଲ଼", w = 1.0 },
    { text = "ଲ", w = 1.15, context = "word_final" },
    { text = "ଲ", w = 1.05, context = "intervocalic" },
    { text = "ଲ", w = 0.02, context = "word_initial" },
]
```

Regenerate and verify with:

```
npx tsx Appv2/scripts/gen-rules.ts
npx tsx Appv2/eval/run.ts
```

Key idea. Weights are **not capped at 1.0**. The engine normalises every candidate path against the best one, so only *ratios* matter — which is why a word-final ◌୍ can be given `w = 1.6` to put it ahead of ଲ଼ outright.

The eval run refuses to report a number against a stale rule file, checks that everyday Odia has not moved, and proves every character in section 6 is reachable by *running the engine* rather than by reading the rule tables.